

DIPAM SEN

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ACADEMIC QUALIFICATIONS

Indian Institute of Technology, Kharagpur

Kharagpur, India

Bachelor of Technology, Computer Science and Engineering (CGPA: 9.86)

2024 – 2028

AWARDS AND ACHIEVEMENTS

- Secured **All India Rank 331** in **Joint Entrance Exam (Advanced) 2024** among 250,000 candidates.
- Secured **All India Rank 220** in **Joint Entrance Exam (Mains) 2024** among 1,200,000 candidates.
- Achieved the title of **Expert** on Codeforces with a **peak rating of 1632** (Handle: *dipamsen*)

PROJECTS

Lumen: Graphics Rendering DSL 🌐 | OCaml, Raylib, FFmpeg | *Prof. Jibesh Patra* *Feb'26 – Apr'26*

- Developed the **formal grammar** and **static type system** for a DSL targeting graphics simulations and visuals.
- Implemented an OCaml **transpiler pipeline** executing **lexical analysis**, **AST parsing**, and **C code generation**.
- Designed AST semantics for **lexical scoping**, **event-driven blocks**, **vector types**, and **rendering primitives**.
- Added backend targets for both **interactive window** environments and **video rendering** pipelines via FFmpeg.

Git Version Control Client 🌐 | C++, Systems Programming, zlib *Apr'26 – May'26*

- Co-developed a Git client, implementing low level **object storage**, content-addressable **hashing**, and **repository cloning**.
- Implemented core utility commands for reading/writing Git's **blob**, **tree**, and **commit** object formats with byte accuracy.
- Supported cloning over HTTPS, including **ref discovery**, **pack-file parsing** and zlib-based decompression of Git objects.
- Used **OpenSSL** for **SHA-1 hashing** to generate content-addressable object identifiers as per Git's internal model.

Raft Distributed Consensus Engine 🌐 | Go, Distributed Systems, Concurrency *May'26 – Jun'26*

- Implemented the Raft consensus algorithm, with **log replication** and **synchronization** across distributed nodes.
- Designed concurrent communication between cluster nodes using **goroutines** and **synchronization primitives**.
- Ensured **safety** and **consistency guarantees** under **node failures**, **network partitions**, and **leader election**.
- Built a replicated state machine with **persistent logs** and **majority-based commits** following the Raft protocol.

CPBattles 🌐 | Node.js, React, MongoDB, PostgreSQL, CI/CD Pipelines *Jun'25 – Jul'25*

- Built a full-stack competitive programming platform with an **Express** backend and **React** frontend for realtime battles.
- Integrated with **Codeforces OAuth 2.0** (via OpenID Connect) for user authentication and account verification.
- Created a **polling-based submission checker** that periodically queries the Codeforces API to update standings.
- Managed scheduled events using **Agenda**, database: **MongoDB**, **PostgreSQL**, and CI/CD pipelines deploying to **Azure**.

IQPS: Intelligent QP Search 🌐 | **metaKGP** | Rust *Jun'25 – Aug'25*

- Maintainer of IQPS, a Rust web app serving **12,000+** **past examination papers** across **2000+** **courses**.
- Integrated **Slack notifications** for paper uploads and imports, enabling real-time monitoring and updates for maintainers.
- Streamlined admin dashboard workflows for approving, editing, replacing and deleting papers, reducing manual overhead.
- Built a **bulk PDF import pipeline** that deduplicates entries via SHA-256 hashing, with transactional rollback on failure.

TECHNICAL SKILLS

- **Programming Languages:** C/C++, Rust, OCaml, Go, Python, JavaScript/TypeScript, Bash
- **Frameworks and Libraries:** React, Express, Node.js, Flask, Next.js, Axum, Tokio
- **Technologies:** Git/GitHub, Docker, Typst, LaTeX, MongoDB, PostgreSQL, GitHub Actions, Linux

COURSEWORK

Programming and Data Structures (Theory and Lab) | Algorithms (Theory and Lab) | Systems Programming Laboratory | Discrete Structures | Probability and Statistics | Signals and Systems | Formal Language and Automata Theory | Software Engineering (Theory and Lab) | Switching Circuits and Logic Design

POSITIONS OF RESPONSIBILITY

- **Maintainer, metaKGP** 🌐 *Feb'26 – Present*
 - Contribute to and maintain **multiple open source projects** under **metaKGP** for the student community.
 - Organised events for familiarising the student community to open source and contributing to open source software.
- **Core Team Member, Kharagpur Open Source Society** *Jul'25 – Present*
 - Conducted **Kharagpur Winter of Code**, a program on open source with **360+** **participants**, **40+** **projects**.
 - Organized hands-on workshops and tutorials to familiarise students with Git/GitHub, Shell, and Linux.